



SYSTEM DESIGN DOCUMENT

TECH 101 ASSIGNMENT · BHANZU

K-12 Learning Platform

Personalisation

The goal is to help users discover relevant courses based on their interests, skill levels, and career goals, leading to increased engagement, completion rates, and overall satisfaction.

SUBMITTED BY-

NISHA VERMA

DOCUMENT TYPE

**System Design Document
v1.0**

PLATFORM

Bhanzu · K-12 Math

DATE

August 2026

Table of Contents

01	Executive Summary	
	Platform context, problem, solution overview	
02	User Personas	
	Learners, Trainers, Platform Admins	
03	User Stories	
	Role-based story mapping from Figma board	
04	Functional Requirements	
	10-feature requirements table with priorities	
05	Prioritisation Matrix	
	Effort vs. Impact quadrant analysis	
06	Non-Functional Requirements	
	Performance, scalability, security & compliance	
07	Success Metrics	
	North Star metric + supporting KPIs	
08	Core System Design	
	Architecture overview and service map	
09	Wireflow	
	Low-fidelity wireframes — Learner, Trainer, Admin	
10	Trade-Off Analysis	
	Personalisation vs. Performance decision	
11	Key Assumptions	
	Scoping and design constraints	
12	Possible Improvements	
	Future extensions and enhancements	
13	Conclusion	
	Summary and next steps	

01 ☆ Executive Summary

PLATFORM

Bhanzu K-12 Math

FOCUS AREA

Personalised Learning

PRIMARY USERS

Learners, Trainers, Admins

SCOPE

UKG – Grade 12 (Math)

Context & Problem

Bhanzu is a K-12 math learning platform that delivers live classes to students from UKG through Grade 12. While the core live-session infrastructure is in place, the platform currently lacks personalisation — every student receives the same post-class experience regardless of their individual performance, pace, or learning gaps.

This results in key pain points: learners who struggle receive no targeted reinforcement; motivated students have no clear visibility into their learning journey; trainers lack a consolidated view of student performance to tailor live guidance; and platform administrators have limited analytics to measure recommendation effectiveness or predict churn signals such as credit exhaustion.

Proposed Solution

This Project specifies a **Personalised Learning Recommendation System** built on top of the existing Bhanzu infrastructure. The system introduces three core capabilities: (1) an adaptive Practice Engine that generates skill-gap-targeted questions post-assignment; (2) a Learning Roadmap giving learners clear visibility into completed, in-progress, and upcoming content; and (3) an On-Demand Doubt Resolution mechanism so students are never blocked waiting for the next live class.

Business Objectives & Success Targets

This system is designed to achieve three measurable business outcomes:

- Engagement +25%: Measured via Weekly Active Practice Rate (WAPR)— target 45% at 6 months, 65% at 12 months
- Completion +30%: Measured via Module Completion Rate — target $\geq 70\%$ module-level progression
- Satisfaction +20%: Measured via Learner Satisfaction Score (post-session CSAT) — target improvement from baseline to +20%

02



User Personas



Learners

UKG - 12th Grade

GOALS

- ✓ Master mathematical concepts at their own pace
- ✓ Understand progress through a clear roadmap
- ✓ Get quick help when stuck during practice

PAIN POINTS

- ✗ No targeted practice after completing assignments
- ✗ Long recordings — no quick revision option
- ✗ Must wait for live class to resolve doubts
- ✗ Motivation drops with no engagement feedback

AGE RANGE

5-18 years

TECH LEVEL

Basic-Intermediate

PLATFORM

Mobile-first



Trainers

Live Session Instructors

GOALS

- ✓ Prepare for sessions quickly and efficiently
- ✓ Understand each learner's history at a glance
- ✓ Keep live class disruptions to a minimum

PAIN POINTS

- ✗ Lengthy prep material with no concise summary
- ✗ No consolidated view of a learner's progress
- ✗ Slow technical support pipeline during live classes
- ✗ Unclear which assignments learners should complete post-class

ROLE

Subject Instructor

TECH LEVEL

Intermediate

PLATFORM

Desktop + Mobile



Platform Admins

Operations & Oversight

GOALS

- ✓ Measure recommendation effectiveness
- ✓ Proactively identify at-risk learners
- ✓ Monitor course quality across the platform

PAIN POINTS

- ✗ No visibility into learner interaction with recommended content
- ✗ Cannot predict credit exhaustion in advance
- ✗ No aggregate view of course quality trends

ROLE

Ops / Business Admin

TECH LEVEL

Intermediate

PLATFORM

Desktop

03 User Stories

Learners UKG - 12th Grade

- Practice & Reinforcement** As a learner, I want additional practice opportunities after completing my assignments so that I can strengthen my understanding and improve my performance.
- Motivation & Engagement** As a learner who struggles with maths, I want to feel motivated to continue learning on the platform so that I stay engaged and consistently complete my learning journey.
- Learning Roadmap** As a learner, I want a clear view of my learning roadmap — all modules and topics — so that I know what I've completed, what I'm learning now, and what comes next.
- Quick Revision** As a learner, I want to revise previously taught concepts quickly so that I can prepare for future classes and assessments without watching long recordings.
- Doubt Solving** As a learner, I want a quick doubt-solving option whenever I get stuck while practicing so that I can keep learning without waiting for the next live class.

Trainers Live Session Instructors

- Session Preparation** As a trainer, I want to quickly understand the teaching material for an upcoming session so that I can prepare effectively without spending excessive time on lengthy instructions.
- Learner History at a Glance** As a trainer, I want to easily view a learner's complete learning history and performance in one place so that I can provide personalised guidance during classes.
- Technical Support** As a trainer, I want a faster way to get technical support when learners face issues during live classes so that problems are resolved quickly with minimal disruption.
- Post-Class Assignments** As a trainer, I want learners to clearly know which assignments to complete after each class so that they can practise covered concepts without confusion.

Platform Admins Operations & Oversight

- Recommendation Visibility** As a platform admin, I want visibility into how learners interact with recommended courses so that I can measure whether personalised recommendations are improving engagement and outcomes.
- Credit Exhaustion Alerts** As a platform admin, I want to identify learners whose remaining credits are likely to be exhausted soon so that the sales team can proactively reach out to parents for timely renewals.
- Course Quality & Trends** As a platform admin, I want to monitor course quality and learner performance trends across the platform so that I can identify areas requiring operational or academic improvements.

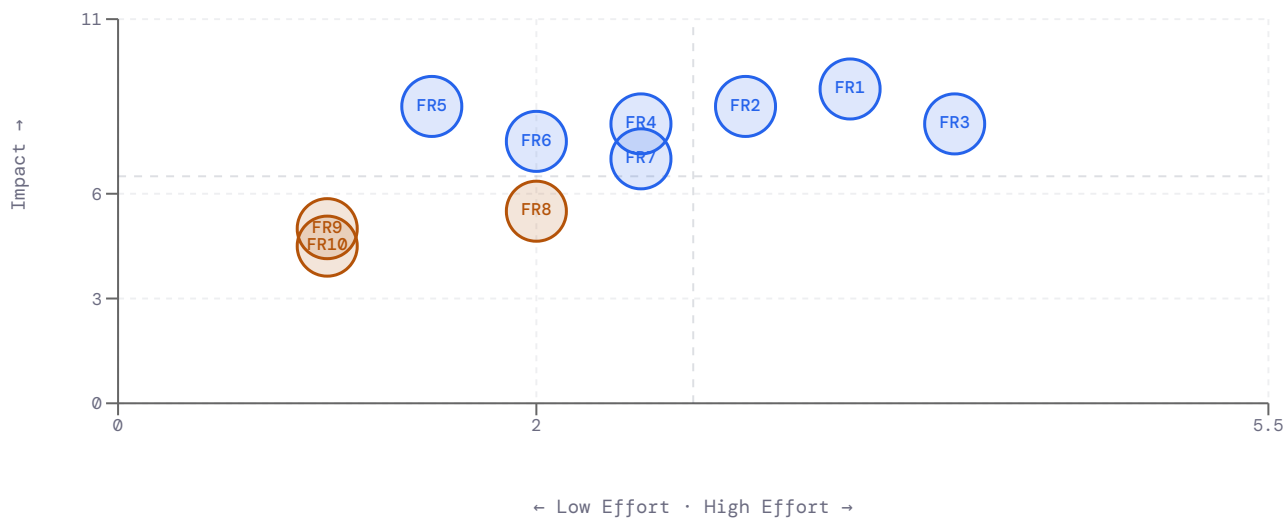
04 Functional Requirements

Personalised Learning Recommendation System — K-12 Platform

#	User Type	Feature	Description	Priority
1	Learner	Adaptive Practice Recommendations	System shall generate additional practice questions/exercises tailored to a learner's completed assignments and skill gaps, to reinforce weak areas post-assignment.	High
2	Learner	Learning Roadmap & Progress Tracking	System shall display a visual roadmap of all modules/topics (completed, in-progress, upcoming) and use progress patterns to trigger motivational nudges for at-risk learners.	High
3	Learner	On-Demand Doubt Resolution	System shall offer an in-platform doubt-solving mechanism (e.g., chatbot, async query, or peer/trainer ping) accessible during self-paced practice, without waiting for the next live class.	High
4	Trainer	Learner Profile & Performance Dashboard	System shall consolidate a learner's complete learning history and performance metrics into a single trainer-facing view to enable personalised guidance.	High
5	Trainer	Post-Class Assignment Visibility	System shall auto-generate and display clear, class-linked assignment lists for learners immediately after each session concludes.	High
6	Admin	Recommendation Engagement Analytics	System shall provide admins a dashboard tracking learner interaction with recommended courses (click-through, completion, drop-off) to evaluate recommendation effectiveness.	High
7	Admin	Platform-Wide Quality & Performance Monitoring	System shall aggregate course quality signals and learner performance trends across the platform to help admins identify areas needing academic or operational intervention.	High
8	Learner	Quick Revision Mode	System shall provide condensed, topic-wise revision content (summaries/flashcards/short clips) as an alternative to re-watching full session recordings.	Medium
9	Trainer	Session Prep Summary	System shall provide trainers a concise, structured summary of teaching material for upcoming sessions, reducing time spent parsing lengthy instructions.	Medium
10	Admin	Credit Depletion Alerts	System shall flag learners approaching credit exhaustion and route this data to the sales team for proactive renewal outreach.	Medium

05 © **Prioritisation Matrix**

Effort vs. Impact analysis. High-impact, low-effort features are quick wins; high-impact, high-effort features are strategic bets.



- Quick Wins**
High impact, low effort — ship first
- FR5 Post-Class Assignments
 - FR9 Session Prep Summary
 - FR10 Credit Depletion Alerts
- Fill-Ins**
Lower impact, low effort — nice to have
- FR8 Quick Revision Mode

- Strategic Bets**
High impact, high effort — core investments
- FR1 Adaptive Practice
 - FR3 Doubt Resolution
 - FR2 Learning Roadmap
- Reconsider**
High effort, lower impact — deprioritise
- (None in current scope)

06 Non-Functional Requirements

1. PERFORMANCE

Quiz Load Time — Quizzes shall load within 2 seconds; if exceeded, the system shall display a loading indicator — the current blank-page delay leaves children confused and causes quiz drop-offs.

Live Session Load Time — Live class interfaces (video, whiteboard, chat) shall load within 3 seconds on a standard broadband/4G connection, since delays directly disrupt teaching time.

2. SCALABILITY

Concurrent User Load — The platform shall support at least 10,000 concurrent live-session users during peak hours (e.g., evening batches) without degradation.

Recommendation Engine Data Growth — As the learner base grows (e.g., 5× in 12 months), the recommendation engine shall scale without a full architecture redesign — more learners means more interaction data, not just more traffic.

3. AVAILABILITY

Uptime SLA — The platform shall maintain 99.9% uptime during scheduled class hours (e.g., 3 PM–9 PM IST), since downtime directly impacts trainers and paying parents.

Graceful Degradation — Core learning features (quizzes, assignments, session access) shall remain available even if the recommendation engine fails — a single component failure must not block platform access.

4. COMPLIANCE

Child Data Protection (COPPA / DPDP Act) — The platform shall comply with India's DPDP Act (and COPPA for international users) for collection, storage, and consent around children's data, including parental consent mechanisms.

Data Localisation — Learner data shall be stored in compliance with applicable data residency requirements (e.g., Indian data stored within India, per RBI/MeitY guidelines).

5. SECURITY & DATA RETENTION

Role-Based Access Control (RBAC) — The system shall enforce RBAC so each role (Learner, Trainer, Admin) can only access data and functions authorised for that role, minimising exposure of sensitive student data.

Data Retention Policy — Learner performance data shall be retained for a defined period (enrollment duration + 2 years) and securely purged afterward, in line with privacy regulations.

6. MONITORING & LOGGING

System Health Monitoring — The platform shall continuously monitor key services (live session servers, recommendation engine, doubt-resolution bot) with automated alerts when error rates or latency cross defined thresholds.

Audit Logging — All access to sensitive learner data (by trainers or admins) shall be logged with timestamps and user IDs, to support accountability and compliance audits.

7. EXTENSIBILITY

Modular Recommendation Logic — The recommendation engine shall be built so new signals (e.g., parent feedback, attendance patterns) can be added without re-architecting the core system.

API-First Design — Core services (learner profile, content catalog, assignment engine) shall expose APIs, enabling future integration with third-party tools (school ERPs, parent apps) without major rework.

07 Success Metrics

NORTH STAR METRIC

Weekly Active Practice Rate (WAPR)

The percentage of enrolled learners who complete at least one personalised practice session per week. This metric captures whether learners are actively using the personalisation system, not just logging in — making it the truest signal of recommendation engine value.

Current Baseline 6-Month Target 12-Month Target

~18% ≥ 45% ≥ 65%

Supporting KPIs

LEARNER ENGAGEMENT

- Recommendation Click-Through Rate** ≥ 40%
Measures relevance of surfaced practice content
- Doubt Resolution Response Time** < 5 min
Ensures learners are unblocked quickly during practice
- Roadmap View-to-Practice Conversion** ≥ 30%
Confirms roadmap drives action, not just awareness

LEARNING OUTCOMES

- Post-Practice Quiz Score Improvement** +15% avg
Validates that adaptive content improves performance
- Module Completion Rate** ≥ 70%
Indicates learners are progressing through curriculum
- Learner Satisfaction Score (CSAT)** ≥ 20%
Measures post-session ratings and engagement sentiment

TRAINER EFFECTIVENESS

- Session Prep Time Reduction** ≥ 30%
Validates Session Prep Summary feature value
- Trainer Learner Profile Utilisation** ≥ 60%
Confirms trainers actively consult learner dashboards

PLATFORM OPERATIONS

- Credit Exhaustion Alert Lead Time** ≥ 7 days
Enables timely sales outreach before credits expire
- Recommendation Engine Uptime** 99.9%
Tracks SLA compliance for core system availability

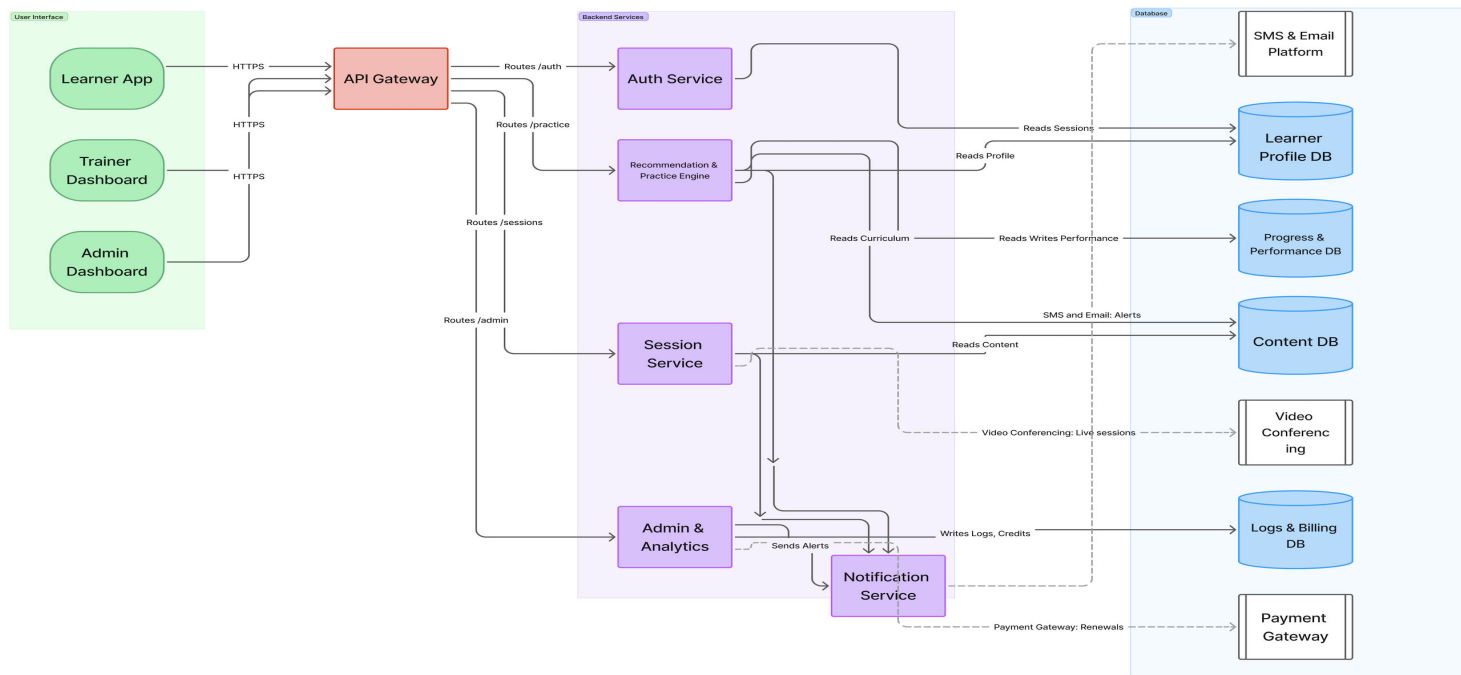
NORTH STAR METRIC

Weekly Active Practice Rate

Target: ≥ 45% at 6 months, ≥ 65% at 12 months (from ~18% baseline)

08 Core System Design

The platform follows a layered, service-oriented architecture. Three user-facing applications communicate through a central API Gateway, which routes requests to dedicated backend services. Each service has narrowly defined responsibilities and communicates with dedicated data stores and third-party integrations.



USER INTERFACE

Learner App (mobile-first), Trainer Dashboard, and Admin Dashboard — all communicating via HTTPS to the API Gateway.

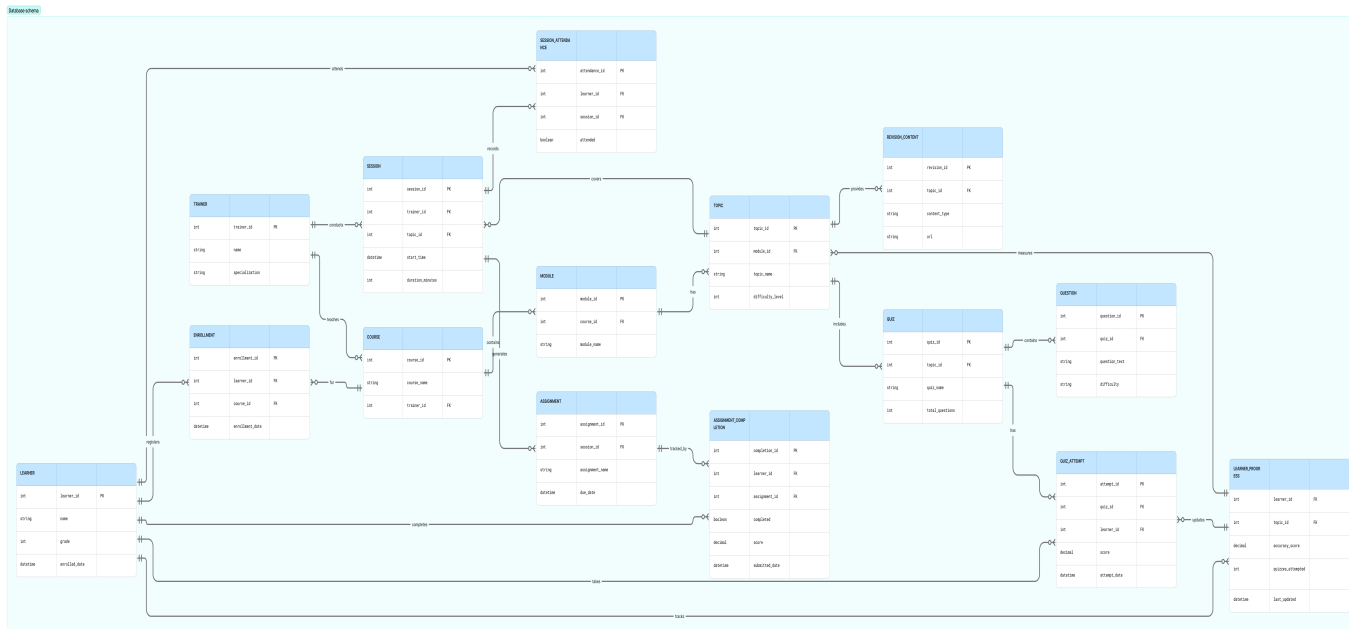
BACKEND SERVICES

Auth Service, Recommendation & Practice Engine, Session Service, Admin & Analytics, and a Notification Service for alerts.

DATA STORES

Dedicated databases for profiles, performance logs, content/curriculum, billing, plus third-party Video Conferencing and Payment Gateway integrations.

Database Schema



09 Wireflow

Low-fidelity wireframes — Learner, Trainer & Admin — key screens mapped to FR/system design (from Figma board).

LEARNER — HOME

Hi, Aarav
Grade 6 · Math

Your Roadmap

Fractions to Decimals — 60% complete

Recommended for you

Practice: Adding Fractions
Based on your last quiz

Practice: Number Line Basics
Quick refresh before class

Revision: Decimals Recap
2 min read

Quick Revision
Review before your next class

?

Home Roadmap Doubts Profile

LEARNER — ROADMAP

Your Roadmap

✓ Whole Numbers
5/5 topics

● Fractions
3/5 topics

● Decimals
1/5 topics

🔒 Percentages
Locked

🔒 Basic Algebra
Locked

Home Roadmap Doubts Profile

TRAINER — DASHBOARD

Bhanzu Trainer

Dashboard

My Learners

Sessions

Assignments

Support

Good evening, Priya

Upcoming Session —
5:00 PM
Fractions · Grade 6B · 12 learners

View Prep Summary

Doubts Waiting
3 learners need a response

Open Doubt Queue

Learner Performance

Learner	Progress	Score	
Aarav K.		91%	View >
Diya S.		74%	View >
Rohan M.		58%	View >
Meera P.		96%	View >

ADMIN — ANALYTICS DASHBOARD

Admin — Platform Overview

Active Learners
12,480

Engagement Rate
+22%

Completion Rate
68%

Avg. Satisfaction
4.3/5

Recommendation Engagement

Mon Tue Wed Thu Fri Sat Sun

Credit Alerts (14)

● Aarav K. 2 left

● Diya S. 1 left

● Rohan M. 3 left

Platform Quality Monitoring

Trainer	Avg Rating	Sessions	Flag
Priya N.	4.7	32	—
Karan S.	3.9	28	Review

- > **Wireflow:** Learner: Home → personalised recommendations → Roadmap for module view → "?" FAB for on-demand doubt resolution. Trainer: Dashboard → session prep summary → learner performance table. Admin: Overview → recommendation engagement analytics → credit alerts → quality monitoring flags.

System Design| Databse schema| wireframe link:

<https://www.figma.com/board/CfoDYsaRyo53mxSGZFeY17/K-12-learning-platform?node-id=0-1&t=TuLEY0uxhYv1AnkE-1>

10 Trade-Off Analysis

PRACTICE ENGINE

Personalisation vs. Performance

Balancing deep customisation with fast response times to optimise K-12 learner engagement.

PERSONALISATION

Deep, real-time analysis

Chosen Balance

PERFORMANCE

Sub-2s response time

01 What's in Conflict

- ✗ **Personalisation needs data:** Every recommendation depends on reading a learner's profile, skill level, and recent performance from DB.
- ✗ **More data, more time:** The richer and more real-time the analysis, the longer it takes to compute — adding visible user latency.
- ✗ **K-12 users don't wait:** For young learners, delays lead directly to abandonment, as seen in quiz white-screen drop-offs.
- ✗ **The core tension:** Accuracy and speed pull directly against each other within the same practice workflow.

02 How We Prioritise

- **Speed wins by default:** A fast, good-enough recommendation seen immediately beats a precise one arriving too late.
- **Attention span constraint:** K-12 users have low lag tolerance; performance directly drives feature engagement.
- **Diminishing returns:** Insights from 5 mins ago vs. 24 hours ago yield minimal accuracy gains relative to compute cost.
- **Conclusion:** Protect the sub-2-second response NFR first; adjust personalisation depth within that boundary.

03 Product Decision

- ✓ **Precompute over live calculation:** Generate learner insights on a scheduled background pipeline rather than on-demand.

- ✓ **API Caching:** Cache frequently accessed results with a freshness window of roughly 15–30 minutes.

- ✓ **Event-driven refresh:** Invalidate cache upon quiz or session completion to keep content relevant when it counts.

WHY STALENESS IS SAFE HERE

A learner's core skill profile does not shift dramatically within 15–30 minutes, making short-term staleness harmless. Tying refreshes directly to completed sessions ensures both high responsiveness and updated recommendations.

FR-01 Adaptive Practice

FR-02 Roadmap

NFR Performance

NFR Scalability

11 ⓘ Key Assumptions

01 Learner Motivation

We're assuming students are interested and motivated enough to actively use the practice system — the Practice Engine's value depends on learners choosing to engage with recommended questions, not just being shown them.

02 Live Session Data Continuity

We're assuming the third-party video conferencing tool used for live sessions supplies continuous, reliable data back to the Session Service — attendance, engagement signals, session logs — without gaps that would break downstream features like Session Prep or Post-Class Assignments.

03 Network Stability

We're assuming learners generally have stable internet access; the performance NFRs (sub-2-second quiz load) assume typical broadband/4G conditions, not poor or intermittent connectivity.

04 Single Admin Persona

We're assuming one internal "Platform Ops/Business Admin" role is sufficient to represent Platform Admin needs, rather than separately designing for School-level Admins or a Super Admin/configuration role.

05 Cache Freshness is Safe

We're assuming a 15–30 minute caching window for practice recommendations doesn't meaningfully hurt personalisation, since a learner's skill level isn't expected to shift significantly within that time.

06 Single-Subject Scope

We're assuming the platform serves Math specifically, not a multi-subject catalog — this shapes the Practice Engine's design as curriculum-sequenced rather than course-discovery based.

07 Parents as a Non-Primary Persona

We're assuming parents/guardians don't need their own dashboard or persona, even though they're implicitly involved via Credit Depletion Alerts and billing.

12 Possible Improvements & Extensions



Gradewise Leaderboard

Introduce a grade-wise leaderboard for quiz scores, giving learners a light competitive/social motivation layer alongside personalised recommendations — this would also help address the learner motivation assumption above by adding an engagement mechanism beyond intrinsic interest.

Phase 2



Recommendation Feedback Loop

Let learners rate or skip recommended practice questions, feeding that signal back into the Practice Engine to improve relevance over time rather than relying purely on performance data.

Phase 2



A/B Testing the Cache Window

Experiment with different cache freshness intervals (e.g., 10 min vs. 30 min) to validate whether the assumed safe staleness window actually holds up against real engagement/completion data.

Phase 1



Parent-Facing View

Extend the platform with a lightweight Parent persona/dashboard — progress summaries, credit/renewal alerts, and session attendance — building on the Notification Service already in the architecture.

Phase 3



Multi-Subject Expansion

Generalise the Practice Engine and Content DB structure to support subjects beyond Math, which would also test whether the current system design holds up at greater content complexity.

Phase 3



Offline / Low-Connectivity Mode

Address the network-stability assumption directly by allowing limited offline access to practice content, syncing progress once connectivity resumes — relevant given India's variable network conditions across regions.

Phase 2

PRIORITISATION NOTE

Phase 1 items can be explored immediately after launch as low-cost experiments. **Phase 2** items require new data signals or modest architectural additions. **Phase 3** items depend on successful multi-cohort rollout and represent significant platform expansions — re-evaluate with live usage data.

13 Conclusion

What This Project Delivers

This document defines a full personalised learning recommendation system for Bhanzu's K-12 math platform — designed to address real friction points across three core user types: learners who lack targeted practice, trainers who lack consolidated student context, and admins who lack visibility into recommendation effectiveness and platform health.

The 10 functional requirements are grounded in specific user stories from Figma. The prioritisation matrix identifies quick wins (Post-Class Assignments, Session Prep Summary) that can ship with minimal effort while building towards the strategic bet: a live Adaptive Practice Engine that meaningfully improves learning outcomes.

Key Design Decisions

- **Performance over precision**
The sub-2-second NFR takes precedence over recommendation depth — speed wins for K-12 users.
- **Precomputed recommendations**
Background pipelines with event-driven cache invalidation balance freshness and latency.
- **Graceful degradation first**
Core learning features remain available even if the recommendation engine fails.
- **API-first architecture**
All core services expose APIs, enabling future integrations without rework.